Fitness Feedback

Knowledge Update

Elite Refinement (KER)

f_1 f_2 f_3 … f_D

P_1 1 0 1 0 1 0 … 1 0 0

P_2 0 1 0 1 1 0 … 0 1 1

P_3 1 1 0 0 1 0 … 1 0 0

⋮

P_N 0 1 1 0 0 1 … 1 1 0

Accumulated Evolutionary Experience

$t-2$ $t-1$ … t

Cumulative Feature-Importance Matrix W

f_1 f_2 f_3 … f_{D-2} f_{D-1} f_D

High W_j (Key features → guide $0 \to 1$) **Low W_j (Redundant → prune $1 \to 0$)**

Elite (before)

0 1 1 0 1 0 0 1 … 0 1 0

Knowledge-guided Directional Flip

1 1 1 0 0 0 0 0 … 1 0

$0 \to 1$ $1 \to 0$

Fitness Re-evaluation

Improved?

Yes No

Accept (refined elite) *Retain (original elite)*

Elite (after)

1 1 1 0 0 0 0 1 … 1 0

Dynamic Memory Decay $(W \leftarrow \gamma W)$

Fitness Evaluation

Closed-loop Knowledge Learning (1-to-1 Replacement)